

ATO Detection via Device Embeddings

Can login device sequences catch account takeover in real time?

Answer: Yes — with the right signal and training configuration.

FastText on structured feature tokens achieves **AUC 0.985** on novel attacks at sub-millisecond inference latency.

Three experiments · Adversarial critique & debate · Peer review · Production analysis

github.com/chris-santiago/ato-device-embeddings

Which Signal Works?

Three experiments compared embedding strategies across attack types.

Signal	Novel AUC	Fleet AUC	Spoof AUC	Verdict
Feature tokens, mean-pool	0.993	0.939	0.818	✔ Recommended
Feature tokens, concat string	0.981	0.933	0.763	✘ Below trivial on spoof
Word2Vec on device IDs	—	0.891	—	✔ Offline fleet detection only
FastText on device IDs	silhouette -0.051	—	—	✘ Destroys cluster structure
Trivial set-membership	0.750	0.750	0.791	Fallback only

Key insight: Mean-pool is the *only* configuration that beats the trivial baseline on spoof attacks — the hardest case, where the attacker matches the victim's OS, browser, and language and differs only on timezone.

Why Mean-Pool Beats Concatenated Strings

Character n-grams in a concat string span feature boundaries (`ari_ut` , `i_utc`), injecting spurious signal that dilutes any single mismatched feature. Mean-pooling embeds each feature independently — a differing timezone contributes its full weight to the score.

Test	Result	Verdict
Bootstrap CIs on all 4 deltas	All exclude zero	✓
Window sweep (w=1 to w=6)	Best concat recovers only 43.6% of gap	✓
Prefixed-concat format	Silhouette gap 0.090 > 0.05 threshold	✓
Tz-position permutation	Every concat ordering below mean-pool spoof AUC	✓
Trivial baseline comparison	Mean-pool +0.027; concat -0.028 over trivial	✓

Score: 7/7 — confirmed across three independent investigations.

! Critical Finding: Training Configuration Matters

An initial investigation reached the **opposite conclusion** — concat beat mean-pool on every metric. Root cause: a degenerate training configuration causes silent within-feature embedding collapse. All timezone values converge to near-identical vectors (sim = 0.9993), eliminating the signal mean-pool depends on.

Configuration	Within-feature sim	Spoof AUC	Conclusion
CBOW + per-event corpus	0.9993 — collapse	0.384 (<i>below chance</i>)	Mean-pool refuted
Skip-gram + per-account corpus	0.392 — healthy	0.818	Mean-pool confirmed

This failure is silent. Novel and fleet AUC look reasonable even under collapse — only a within-feature token similarity check (T8) detects it. Run after every retraining cycle; alert if similarity > 0.5.

Recommended Production Architecture

```
At login (<1ms):
tokens      = [os_X, br_X, tz_X, lang_X, net_X, sc_X]
embedding   = mean(fasttext[t] for t in tokens)
risk_score  = cosine_distance(embedding, account_centroid)
→ step-up auth if risk_score > threshold

Monthly retraining:
corpus: one sentence per account (all login events concatenated)
FastText: sg=1, epochs=20, negative=10, window=6, vector_size=64
✓ Health check: within-feature cosine sim – halt deployment if > 0.5

Offline (daily/weekly):
device_id_history → Word2Vec centroid → fleet/reuse review queue

Fallback (<20 events or service unavailable):
per-account known-device hash set → step-up if unseen profile
```

Risk to monitor: The mean-pool spoof advantage over the trivial baseline is +0.027 — statistically significant but operationally narrow. Include spoof-specific AUC in any A/B evaluation before cutover.